

SOUFYAN RACHDI

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GitHub | LinkedIn | Portfolio

SUMMARY

Full Stack & Mobile Developer building real-time, cross-platform apps with Flutter, Node.js and NestJS. Shipped products from BLE health monitoring to live multiplayer games; now pursuing a research Master's in AI at ISIMG.

EXPERIENCE

Software Engineering Intern | Caredify

Jan 2026 – Jun 2026

Built the Flutter patient app and backend APIs for an AI-assisted remote ECG monitoring platform.

Competitive Programming Manager | CPC ISIMG

Sep 2024 – Feb 2026

Organized programming workshops and contests, and mentored students in algorithms and problem-solving.

PROJECTS

Caredify — AI-Assisted Remote ECG Monitoring Platform

Feb 2026 – Jun 2026

Flutter • Kotlin • Node.js • Express.js • MongoDB • Python/TensorFlow • BLE • Render

- Worked in a team on an end-to-end cardiac monitoring system: wearable Movesense ECG sensor, Flutter patient app, Node.js backend (32 endpoints, 6 collections), AI classifier, and cardiologist dashboard.
- Built the Flutter app with BLE streaming from the sensor, live ECG waveform, background monitoring (Android foreground service), GPS map, and nearby hospital search.
- Built the backend that sends ECG data to a Python AI service (ResNet-1D model, 5 arrhythmia classes, 98.73% test accuracy on the MIT-BIH dataset).
- Implemented the emergency flow: an alert only fires after 3 consecutive dangerous readings (to avoid false alarms), then a 60-second cardiologist review, then an SMS with GPS location to emergency contacts.

Enkivo — Real-Time Multiplayer Quiz Platform

Jul 2026 – now

Flutter • NestJS • Next.js • Socket.IO • Redis • Prisma • Supabase • Render • Cloudflare

- Building a multiplayer quiz game with 5 modes (solo, casual, ranked, battle royale up to 10 players, private rooms) on mobile (25 screens), web, and an admin dashboard.
- Built a server-controlled game loop with Socket.IO rooms and Redis: the server handles questions, timing and scoring, so players can't cheat.
- Designed matchmaking with atomic Redis scripts, automatic mid-game reconnection, and cleanup of abandoned matches.
- Backend with 66 REST endpoints and a 26-model database; includes an in-game economy, leaderboards, friends, and support for 26 languages.

Football Manager — Multiplayer Football Management Game

Aug 2026 – now

Flutter (BLoC) • NestJS • Socket.IO • Redis • BullMQ • Prisma • CockroachDB • Render

- Building a real-time multiplayer football manager: 8-player lobbies (AI bots fill empty spots), a live transfer auction, and 28-match round-robin leagues.
- Built the auction engine with atomic Redis Lua scripts: automatic refunds for outbid players, anti-snipe timer extensions, and no race conditions between bidders.
- Built a match simulation engine (Poisson-based) that uses ratings, morale, fatigue, chemistry and tactics, streamed live minute by minute over WebSockets.
- Added self-recovery: the server restores unfinished leagues after a restart, and a watchdog unfreezes stuck phases. Also enforced single-device login. 79 automated tests.

PROJECTS (Continued)

KabaadHub — Recyclable Materials Marketplace (freelance)

Jul 2025 – Sep 2025

Flutter • Riverpod • Firebase • Cloud Functions • Google AdMob

- Built a scrap marketplace app (36 screens) with listings, chat, push notifications, and search sorted by distance.
- Added sign-in with email, phone OTP, Google, and guest mode, plus admin moderation of every new listing.
- Wrote 15 Firebase Cloud Functions and security rules for notifications, account deletion, and user data protection.
- Built a Flutter Web admin dashboard (users, moderation, analytics, broadcasts) and AdMob ads that can be switched remotely without a new app release.

Mkarchi — Multi-Platform Developer Tooling Ecosystem

2025 – Present

Python • PyPI • Electron • Next.js • AST Parsing • TypeScript • Tailwind CSS

- Authored and published an open-source Python CLI package on PyPI to automate project scaffolding and repository architecture reverse-engineering from concise tree definitions.
- Engineered an Electron desktop visual workspace for blueprint editing alongside the official Next.js web portal at mkarchi.me.

EDUCATION

Master of Science in Computer Science & Multimedia (Research Track in AI) | ISIMG

2026 – Present

Research-track program focused on AI: machine learning, multimedia data processing (image and video), natural language processing, data mining and fusion, and biometric signal processing.

Bachelor's Degree in Computer Science (Licence Informatique) | ISIMG

2025 – 2026

Specialization in software engineering, distributed architectures, and databases. Graduation project (PFE): Caredify, an AI-assisted remote ECG telemetry platform.

Integrated Preparatory Cycle (MPI: Math, Physics & Computer Science) | ISIMG

2023 – 2025

Two years of intensive foundational studies in advanced mathematics, physics, data structures, and algorithmic complexity.

Baccalaureate in Computer Science | Lycée Route de Gabès (LRG)

2022 – 2023

Graduated with Good (Bien) honors (Moyenne 15.79/20).

SKILLS

- **Languages:** Python • Dart • TypeScript • JavaScript • C++ • Java
- **Frameworks:** Flutter • NestJS • Node.js/Express • Next.js • React • Riverpod • BLoC
- **Real-time & Data:** Socket.IO • Redis • BullMQ • Prisma • PostgreSQL/CockroachDB • MongoDB • Firebase • Supabase
- **AI & Hardware:** TensorFlow/Keras • BLE • Kotlin
- **Tools:** Git • GitHub • Docker
- **Spoken languages:** English, French, Arabic
- **Certifications:** CodinGame C++, HackerRank Problem Solving
- **Awards:** 1st Place Winner — ISIMG Local Programming Contest 1.0 (Team CPC0, 2024)